

ANNAPURNA

Project Name: Annapurna- Home-made Healthy Food Delivery System

Problem Statement:

Many people such as students, working professionals, elderly citizens, patients, and busy families, pregnant ladies, are unable to prepare home process healthy meals regularly due to lack of time, physical limitations, or work commitments. Existing food delivery platforms mainly focus on restaurant food, which often contains excessive oil, spices, preservatives, and unhealthy ingredients.

There is a need for a system that provides affordable, hygienic, clean, and customizable meals prepared by trained food processing staff and delivered according to users' health requirements and preferences.

Why does this problem exist?

- Busy lifestyles reduce cooking time.
 - Increase in unhealthy eating habits.
 - Lack of affordable healthy food options.
 - Limited personalized nutrition services.
 - Growing health problems such as obesity, diabetes, and hypertension.
 - Restaurant food often contains excessive oil and spices.
 - Elderly citizens and patients require special diets.
 - Working families struggle to maintain balanced meals.
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Who are the stakeholders?

Internal Users

- System Administrator
- Food Processing Staff (House wife`s and Member`s)
- Marketing system management
- Dietitian Expert

- Delivery Personnel
 - Customer Support Staff
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External Users

- Customers
 - Families
 - Students
 - Working Professionals
 - Elderly Citizens
 - Patients
 - Corporate Organizations
 - Food Suppliers
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Expected System Users:

- Individual Customers
 - Families
 - Corporate Employees
 - Fitness Enthusiasts
 - Patients
 - Senior Citizens
 - Students
 - Healthcare Institutions
 - Work site plants workers, staff and other sites
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Objectives of the Project

1. Provide home-made healthy and hygienic food.
2. Enable personalized meal customization.

3. Control spice and oil levels according to user preferences.
 4. Deliver affordable meals to all sections of society.
 5. Improve public health through better hygiene safe food.
 6. Reduce dependency on unhealthy restaurant food.
 7. Support elderly citizens and patients with special diet plans.
 8. Create employment opportunities for food processing staff.
 9. Minimize food waste through demand prediction.
 10. Build a scalable national healthy food ecosystem.
 11. It will provide the house wife`s and house member`s their income source
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Expected Outputs

- Personalized meal recommendations.
 - Customized food delivery services.
 - Cleanness and hygiene report tracking reports.
 - User health profile management.
 - Kitchen performance reports.
 - Delivery tracking information.
 - Customer feedback reports.
 - Business analytics dashboards.
 - Subscription management reports.
 - Food quality and hygiene monitoring reports.
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Hands-on Activity:

Identify Strategic, Tactical and Operational Information:

Operational Information:

- Daily order processing.
- Food preparation schedules.
- Delivery assignment.

- Payment processing.
- Customer support activities
- Daily inventory usage

Strategic Information:

- National healthy food demand trends.
- Expansion opportunities in new cities.
- Long-term customer health patterns.
- Revenue growth forecasting.
- Market share analysis.
- Customer retention strategies.

Tactical Information:

- Monthly staff allocation.
- Kitchen resource planning.
- Inventory management.
- Subscription management.
- Marketing campaign planning.
- Delivery route optimization.

Use case Activity:

1.Customer Use Case:

Use Case	Actor	Objective
Register Account	Customer	Create account
Login	Customer	Access system
Manage Profile	Customer	Update profile
Browse Menu	Customer	View food items
Search Food	Customer	Find food items
Add to Cart	Customer	Store selected items
Place Order	Customer	Order food
Make Payment	Customer	Complete payment
Track Order	Customer	Monitor delivery
View Order History	Customer	Review previous orders
Cancel Order	Customer	Cancel active order
Submit Feedback	Customer	Provide review

2. System Administrator Use Case:

Use Case	Actor	Objective
Manage Customers	System Administrator	Control user accounts
Manage Menu	System Administrator	Maintain food items
Manage Category (customized Food & Diet Plan)	System Administrator	Organize menu
Manage Orders	System Administrator	Monitor orders
Manage Home Food Makers	System Administrator	View order Prepare Food
Manage Delivery Staff	System Administrator	Control deliveries
Manage Complaints	System Administrator	Resolve issues
Manage Offers	System Administrator	Control promotions
Manage Subscriptions	System Administrator	Maintain plans
Generate Reports	System Administrator	Analyse business data
Monitor System	System Administrator	Ensure system operation
Manage Security	System Administrator	Protect data

3.Dietitian Use Case:

Use Case	Actor	Objective
Create Diet Plan	Dietitian	Prepare meal plans
Review Customer Request	Dietitian	Analyse requirements
Recommend Meals	Dietitian	Suggest suitable food
Approve Special Meals	Dietitian	Validate meal requests
Manage Diet Categories	Dietitian	Organize diet plans
Update Meal Guidelines	Dietitian	Maintain standards
Generate Diet Reports	Dietitian	Monitor diet activities

4. Home Food Makers Use Case:

Use Case	Actor	Objective
View Assigned Orders	Food Processing Staff	Access work orders
Prepare Food	Food Processing Staff	Cook meals
Verify Ingredients	Food Processing Staff	Ensure quality
Update Preparation Status	Food Processing Staff	Update progress
Package Food	Food Processing Staff	Prepare delivery package
Mark Order Ready	Food Processing Staff	Notify completion

5. Delivery Staff Use Case:

Use Case	Actor	Objective
View Delivery Tasks	Delivery Staff	Access deliveries
Accept Delivery	Delivery Staff	Confirm assignment
Pick Up Order	Delivery Staff	Collect food
Deliver Order	Delivery Staff	Deliver food
Update Delivery Status	Delivery Staff	Track delivery
Confirm Delivery	Delivery Staff	Complete delivery

Functional Requirements

1. User registration and secure login.
2. Customer profile and address management.
3. Home food Maker's registration and verification.
4. Manage food menu (add, edit, delete, update availability).
5. Customize food options (portion, spice level, add-ons).
6. Browse and search home-cooked meals.
7. Shopping cart and order placement.
8. Secure online payment and Cash on Delivery.
9. Order tracking and delivery management.
10. Manage subscriptions, offers, and coupons.
11. Complaint and feedback management.
12. Admin management of users, home Makers, delivery partners, reports, and system security.

Non-Functional Requirements

1. Fast response time (within 3 seconds).
2. Secure authentication and encrypted passwords.
3. 24×7 system availability.
4. Reliable and accurate order processing.
5. User-friendly and responsive interface.
6. Scalable to support more users and home chefs.
7. Easy maintenance and future upgrades.
8. Compatible with Android, web browsers, and multiple devices.
9. Data integrity with regular backup and recovery.
10. Privacy and protection of customer and payment information.

Data Type Identification& Soures:

Customer

Internal Data Sources

Source	Data Collected	Purpose
Customer Registration	Name, Mobile, Email, Address	Create customer account
Customer Orders	Food items, Quantity, Order History	Process orders
Customer Feedback	Ratings, Reviews	Improve service

External Data Sources

Source	Data Collected	Purpose
Payment Gateway	Payment Status	Verify payment
GPS/Maps	Delivery Location	Track delivery

Data Type Identification:

Data Item	Type
Customer Name	Structured
Delivery Address	Structured
Reviews	Unstructured
Profile Image	Semi-structured

Home food Makers:

Internal Data Sources

Source	Data Collected	Purpose
Chef Registration	Name, Home Makers Details	Register home chef
Food Menu	Food Name, Price	Manage menu
Orders	Order Details	Prepare food

External Data Sources

Source	Data Collected	Purpose
Food Safety Documents	Verification Details	Verify chef
Bank Details	Payment Information	Chef payouts

Data Type Identification

Data Item	Type
Chef Name	Structured
Food Details	Structured
Food Images	Semi-structured
Kitchen Description	Unstructured

Delivery Partner

Internal Data Sources

Source	Data Collected	Purpose
Delivery Registration	Name, Mobile	Create account
Delivery Orders	Order ID, Status	Deliver food
Delivery History	Completed Deliveries	Track performance

External Data Sources

Source	Data Collected	Purpose
GPS Service	Live Location	Route tracking
Maps	Distance & Route	Navigation

Data Type Identification

Data Item	Type
Delivery Partner Name	Structured
Delivery Status	Structured
Location Coordinates	Semi-structured
Delivery Notes	Unstructured

Administrator

Internal Data Sources

Source	Data Collected	Purpose
User Management	Customer & Chef Data	Manage users
Order Management	Order Details	Monitor orders
Reports	Sales & Revenue	Business analysis

External Data Sources

Source	Data Collected	Purpose
Payment Gateway	Transaction Status	Verify payments
Email/SMS Service	Notification Status	Send alerts

Data Type Identification

Data Item	Type
User Details	Structured
Sales Report	Structured
Complaint Description	Unstructured
System Logs	Semi-structured

Dietitian

Internal Data Sources

Source	Data Collected	Purpose
Dietitian Profile	Name, Qualification, Specialization	Manage dietitian account
Consultation Records	Customer details, Diet plans	Provide personalized nutrition guidance
Nutrition Plans	Meal plans, Calories, Nutritional values	Recommend healthy meals

External Data Sources

Source	Data Collected	Purpose
Medical Reports	Health conditions, Allergies	Prepare personalized diet plans
Nutrition Database	Calories, Nutrients, Food values	Ensure accurate nutritional recommendations

Data Type Identification

Data Item	Type
Dietitian Name	Structured
Qualification	Structured
Diet Plan	Semi-structured
Consultation Notes	Unstructured
Nutrition Chart	Structured
Medical Report	Semi-structured